

KINEMATICS

MULTIPLE CHOICE QUESTIONS

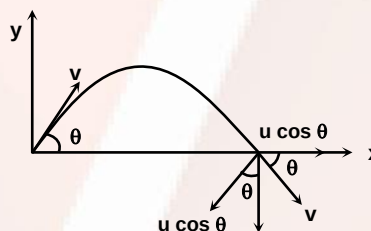
1. A particle is projected upward with an initial speed u making an angle θ with the horizontal. The radius of curvature of the curve at the instant when the particle strikes the ground is

- (A) $\frac{u^2 \cos^2 \theta}{g}$ (B) $\frac{u^2 \sin^2 \theta}{g}$
 (C) $\frac{u^2}{g \sin \theta}$ (D) $\frac{u^2}{g \cos \theta}$

Sol. **D**

Radius of curvature $R =$

$$\frac{v^2}{a_n} = \frac{v^2}{g \cos \theta}$$



2. The co-ordinates of a moving particle at any time t are given by $x = ct^2$ and $y = bt^2$. The speed of the particle is given by

- (A) $2t(c + b)$ (B) $2t \sqrt{c^2 - b^2}$
 (C) $t \sqrt{c^2 + b^2}$ (D) $2t \sqrt{c^2 + b^2}$

Sol. **D**

$$X = ct^2 \quad \text{and} \quad y = bt^2$$

$$\Rightarrow \frac{dx}{dt} = 2ct \quad \& \quad \frac{dy}{dt} = 2bt$$

$$\text{Since } v = \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$\Rightarrow v = \sqrt{(2ct)^2 + (2bt)^2} = 2t \sqrt{b^2 + c^2}.$$

3. A bus is moving with a constant velocity v_1 along a horizontal road. A man standing inside the bus throws a ball upward with velocity v_0 and catches the ball. The magnitude of displacement of the ball with respect to ground is
- (A) $\frac{v_0^2}{g}$ (B) zero
(C) $\frac{2v_0v_1}{g}$ (D) $\frac{v_0v_1}{g}$

Sol. **C**

$$T = \frac{2v_0}{g}$$

$$X = \frac{2v_0v_1}{g}$$

4. A particle is projected vertically upwards and it attains maximum height H . If the ratio of times to attain height h ($< H$) is $1/3$, then h equals
- (A) $(2/3) H$ (B) $(3/4) H$
(C) $(4/3) H$ (D) $(3/2) H$

Sol. **B**

Let the particle be projected vertically upward with velocity u . Further, let t be the time when projectile be at height h . Then

$$h = ut - \frac{1}{2}gt^2 \quad \text{or} \quad \frac{1}{2}gt^2 - ut + h = 0$$

$$\therefore t^2 - \frac{2u}{g}t + \frac{2h}{g} = 0 \quad \dots (1)$$

From equation (1),

$$\text{Sum of roots, } t_1 + t_2 = -\frac{b}{a} = \frac{2u}{g} \quad \dots (2)$$

Product of roots, $t_1 t_2 = \frac{c}{a} = \frac{2h}{g} \quad \dots (3)$

Given that $\frac{t_1}{t_2} = \frac{1}{3}$ or $t_2 = 3t_1 \quad \dots (4)$

From equations (2) and (4), we get

$$4t_1 = \frac{2u}{g} \text{ or } t_1 = \frac{u}{2g} \quad \dots (5)$$

From equations (3) and (4), we get

$$3t_1^2 = \frac{2h}{g} \text{ or } 3\left(\frac{u}{2g}\right)^2 = \frac{2h}{g}$$

$$\text{or } \frac{u^2}{2g} = \frac{4h}{3} \quad \dots (6)$$

Also, maximum height, $H = \frac{u^2}{2g} \quad \dots (7)$

From Eqs. (6) and (7), we obtain

$$H = \frac{4h}{3} \Rightarrow h = \frac{3H}{4}$$

5. A body A is thrown vertically upward with initial velocity v_1 . Another body B is dropped from a height h at time $t = 0$. Which equation represents correct relationship between distance of separation between the two particles (x) and time t (before they collide with ground)?

(A) $X = |h - v_1 t|$

(B) $X = |h + v_1 t|$

(C) $X = |v_1 t + \frac{1}{2}gt^2|$

(D) $X = |\frac{1}{2}gt^2 - v_1 t|$

Sol. A

$$x_A \uparrow = v_1 t - \frac{1}{2}gt^2$$

$$x_B \uparrow = h - \frac{1}{2}gt^2$$

$$\vec{x}_A - \vec{x}_B = v_1 t - h$$

$$X = |h - v_1 t|$$

6. A man can swim at a speed of 5 km/h w.r.t. water. He wants to cross a 1.5 km wide river flowing at 3 km/h. He keeps himself always at an angle of 60° with the flow direction while swimming. The time taken by him to cross the river will be
- (A) 0.25 hr (B) 0.35 hr
(C) 0.45 hr (D) 0.55 hr

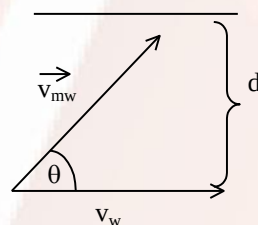
Sol. **B**

The angle between $\vec{v}_{mw}$ and $\vec{v}_w$ is θ (say).

$\Rightarrow$ The required time for crossing,

$$t = \frac{d}{v_{mw} \sin \theta}$$

$$\Rightarrow t = \frac{1.5}{5 \sin 60^\circ} = \frac{3}{5\sqrt{3}} \text{ hr} = \frac{\sqrt{3}}{5} \text{ hr}$$



7. A disc of radius R is rotating inside a room. A boy standing near the rim of the disc, finds that water droplets falling from the ceiling always hit on his head. As one drop hits his head, the next one starts from the ceiling. If height of the roof above his head is H , angular velocity of the disc is
- (A) $\pi \sqrt{\frac{2gR}{H^2}}$ (B) $\pi \sqrt{\frac{2gH}{R^2}}$
(C) $\pi \sqrt{\frac{2g}{H}}$ (D) $2\pi \sqrt{\frac{2g}{H}}$

Sol. **C**

The time period of revolution of the disc is equal to the time of motion of water drop in air.

$$\Rightarrow 2\pi R / \omega = \sqrt{\frac{2H}{g}} \quad \text{or} \quad \omega = 2\pi R \sqrt{\frac{g}{2H}}$$

8. A body is moving in a circle at a uniform speed v . The magnitude of the change in velocity when the radius vector describes an angle of 60° is

- (A) v (B) $v/2$
(C) $\frac{\sqrt{3}}{2}v$ (D) v

Sol. **D**

$$\Delta \vec{v} = \vec{v}_2 - \vec{v}_1$$

$$(\Delta v)^2 = v_1^2 + v_2^2 - 2v_1v_2 \cos \theta = v^2 + v^2 - \frac{2v^2}{2} = v^2$$

$$\Rightarrow \Delta v = v$$

9. A projectile is thrown with a velocity u_0 at an angle θ with the horizontal. The ratio of the rate of change of speed w.r.t. time at the highest point to that at the point of projection is

- (A) $g \sin \theta$ (B) $-g \sin \theta$
(C) zero (D) g

Sol. **C**

$$\frac{dv}{dt} (\text{at the top}) = 0$$

As tangential acceleration at the top = 0

$$\frac{dv}{dt} (\text{at the starting point}) = -g \sin \theta$$

So, required ratio = zero.

10. A particle starts sliding down a frictionless inclined plane. If S_n is the distance travelled by it from time $t = (n - 1)$ sec to $t = n$ sec, then the ratio S_n/S_{n+1} is

(A) $\frac{2n-1}{2n+1}$

(B) $\frac{2n+1}{2n}$

(C) $\frac{2n}{2n+1}$

(D) $\frac{2n+1}{2n-1}$

Sol. **A**

$$S_n = u + (a/2)(2n - 1)$$

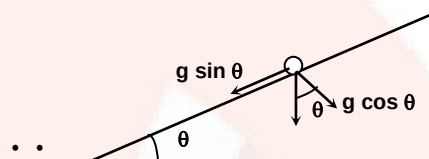
$$S_n = \frac{g \sin \theta}{2}(2n - 1)$$

.. (1)

$$S_{n+1} = \frac{g \sin \theta}{2}(2n + 1)$$

... (2)

$$\text{so, } \frac{S_n}{S_{n+1}} = \frac{2n-1}{2n+1}$$



11. A particle starts moving from rest along a circle of radius 4 m with a constant tangential acceleration of 3 m/sec². The total acceleration of the particle at $t = 2$ seconds makes an angle θ with the tangential acceleration, where θ is equal to

(A) $\tan^{-1}\left(\frac{1}{3}\right)$

(B) $\tan^{-1}(3)$

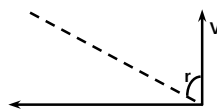
(C) $\tan^{-1}\left(3\frac{1}{3}\right)$

(D) $\tan^{-1}\left(\frac{1}{\sqrt{10}}\right)$

Sol. **B**

At $t = 2$ seconds,

$$v = 6 \text{ m/sec and } a = 3 \text{ m/sec}^2$$



$$\frac{v^2}{R} = \frac{36}{4} = 9 \text{ m/sec}^2$$

$$\alpha = \tan^{-1}(3)$$

12. A particle starts moving with constant acceleration and covers a distance x in first t seconds. The distance travelled by it in the next $2t$ seconds is

(A) $8x$

(B) $3x$

(C) x

(D) none of these

Sol. A

$$x_A \uparrow = v_1 t - \frac{1}{2} g t^2$$

$$x_B \uparrow = h - \frac{1}{2} g t^2$$

$$\bar{x}_A - \bar{x}_B = v_1 t - h$$

$$X = |h - v_1 t|$$

13. A particle moves in the x - y plane with a velocity $v_x = 8t - 2$ and $v_y = 2$. If it passes through the point $x = 14$ and $y = 4$ at $t = 2$ s, the equation of the path is

(A) $x = y^2 - y + 2$

(B) $x = y + 2$

(C) $x = y^2 + 2$

(D) $x = y^2 + y + 2$

Sol. **A**

$$\frac{dx}{dt} = 8t - 2 \Rightarrow \int dx = \int (8t - 2) dt$$

$$\Rightarrow x = 4t^2 - 2t + c$$

$$\text{At } t = 2, x = 14 \Rightarrow C = 2$$

$$x = 4t^2 - 2t + 2 \quad \dots(1)$$

$$\frac{dy}{dt} = 2 \Rightarrow \int dy = \int 2 dt$$

$$\Rightarrow y = 2t + c'$$

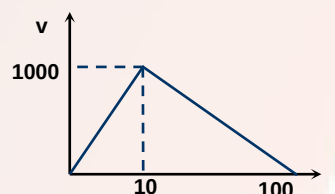
$$\text{At } t = 2, y = 4 \Rightarrow c' = 0$$

$$\Rightarrow y = 2t \quad \dots(2)$$

From (1) and (2), eliminating t

$$x = y^2 - y + 2$$

14. A rocket is fired upwards and its fuel is exhausted in 10 sec. The figure shows its $v-t$ graph, where v is in m/s and t is in sec. What is the maximum height reached?



- (A) 1000×5 m (B) 1000×10 m
(C) 1000×50 m (D) 1000×100 m

Sol. **C**

Maximum height reached = area of $v-t$ graph = $\frac{1}{2} \times 100 \times 1000$ m.

15. If the range of a gun which fires a shell with muzzle speed v is R , then the angle of elevation of the gun is

(A) $\cos^{-1}\left(\frac{v^2}{gR}\right)$

(B) $\cos^{-1}\left(\frac{gR}{v^2}\right)$

(C) $\frac{1}{2}\sin^{-1}\left(\frac{v^2}{gR}\right)$

(D) $\frac{1}{2}\sin^{-1}\left(\frac{gR}{v^2}\right)$

Sol. **D**

$$R = \frac{v^2 \sin 2\theta}{g}$$

$$\sin 2\theta = Rg/v^2$$

$$\theta = \frac{1}{2}\sin^{-1}\left(\frac{Rg}{v^2}\right)$$

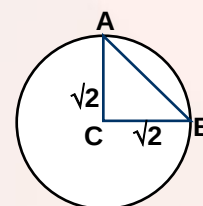
16. A man starts running with constant speed along a circular path of radius $\sqrt{2}$ m. He completes two rounds in 16 seconds. The magnitude of the average velocity in first 2 seconds is

(A) 2 m/s

(B) 1 m/s

(C) 3 m/s

(D) 6 m/s



Sol. **B**

$$a = -kv$$

$$v \frac{dv}{dx} = -kv$$

$$\int_{v_0}^v dv = -k \int_0^L dx$$

$$\Rightarrow v = v_0 - kL$$

17. A particle is moving on a circle of radius R such that at every instant the tangential and radial accelerations are equal in magnitude. If the velocity of the particle be v_0 at $t = 0$, the time for the completion of half of first revolution will be
- (A) $\frac{R}{v_0}$ (B) $\frac{R}{v_0}(1-e^{-\pi})$
 (C) $\frac{R}{v_0}e^{-\pi}$ (D) $\frac{R}{v_0}(1-e^{-2\pi})$

Sol. B

$$\omega^2 R = R \frac{d\omega}{dt}, \quad \omega = \frac{d\theta}{dt} = \frac{\omega_0}{1-\omega_0 t}$$

$$\Rightarrow T = \frac{R}{v_0}(1-e^{-\pi})$$

18. A boy throws a ball in a train, moving at a speed of 18 kmph, with a velocity of $15\sqrt{2}$ m/s at an angle of 45° to the horizontal in the direction of motion of the train. The angle with which it strikes the plane of projection w.r.t. an observer watching from the road nearby to the train is
- (A) 45° (B) 60°
 (C) 37° (D) none of these

Sol. C

$$\text{Time of flight } t = \frac{2u \sin \theta}{g} = 3 \text{ sec}$$

Final velocities w.r.t. ground are

$$v_y = u \sin \theta - gt = 15\sqrt{2} \sin 45^\circ - 10(3) = -15 \text{ m/s}$$

$$v_x = u \cos \theta + u_{\text{train}} = 15\sqrt{2} \cos 45^\circ + 18 \times \frac{5}{18} = 20 \text{ m/s}$$

$$\Rightarrow \tan \theta = \frac{v_y}{v_x} \Rightarrow \theta = 37^\circ$$

19. If position vector of a particle at time t is $\vec{r} = 3t\hat{i} + (8 - 4t)\hat{j}$ m, then
- Particle is moving with constant velocity in straight line
 - Particle is moving with variable velocity in straight line
 - Particle moving is in parabolic path
 - Particle is in rest

Ans. a)

Explanation:

$$\vec{r} = 3t\hat{i} + (8 - 4t)\hat{j} \text{ m}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = 3\hat{i} + 4\hat{j} \text{ m/s}$$

$$\vec{v} = \text{constant}$$

20. A ball thrown in air follows a path given by $y = \frac{x}{\sqrt{3}} - \frac{3g}{8}x^2$ m, where x-axis is taken along the horizontal and y-axis along the vertical. The maximum displacement of the ball along x-direction for which displacement along y is zero equals to

(A) $15\sqrt{3}$ m

(B) $(4/15\sqrt{3})$ m

(C) $4/3$ m

(D) data insufficient

Sol. **B**

On comparing the given equation with the standard equation of a projectile

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$\theta = 30^\circ \text{ and } u = 4/3 \text{ m/s}$$

$$\therefore R = \frac{u^2 \sin 2\theta}{g} = \frac{(4/3)^2 \sin 2(30^\circ)}{10} = \frac{4}{15\sqrt{3}} \text{ m}$$

21. The velocity vector of a particle moving over a horizontal plane is given by $\vec{v} = 2t\hat{i}$. The ratio of the magnitude of instantaneous acceleration of the particle at $t = 2$ sec to the magnitude of displacement of the particle during the interval of first 2 sec is

(A) $3/4$

(B) $4/3$

(C) $5/4$

(D) cannot be determined

Sol. **A**

$$\vec{v} = 2t^2\hat{i}$$

$$\vec{a} = \left. \frac{d\vec{v}}{dt} \right|_{t=2} = 4\hat{i}$$

$$\Rightarrow \vec{a} = 4$$

$$\vec{d} = \int_0^2 2t^2 dt \hat{i} = \frac{2 \times}{4} \hat{i} = \frac{16}{3} \hat{i}$$

22. A particle under the action of a constant force moves from rest upto 20 seconds. If distance covered in first 10 seconds is S_1 and that covered in next 10 seconds in S_2 , then

(a) $S_1 = S_2$ (b) $S_2 = 3S_1$

(c) $S_2 = 2S_1$ (d) $S_2 = 4S_1$

Sol. (b)

$$S = ut + \frac{1}{2}at^2, \text{ given } u = 0, t = 10 \text{ for } S_1$$

$$t = 20 \text{ for } (S_1 + S_2) S_1 = \frac{1}{2}a(10)^2 = 50a$$

$$S_1 + S_2 = \frac{1}{2}a(20)^2 = 200a$$

$$\therefore S_2 = 200a - 50a = 150a - 3S_1$$

23. A simple pendulum hangs from the roof of train. The string is inclined towards the rear of the train. What is the nature of motion of the train?

- (a) Uniform (b) Accelerated
(c) Retarded (d) At rest

Sol. (b)

Train is accelerating horizontally. So, the resultant of horizontal acceleration and vertical acceleration is inclined to the vertical.

24. A juggler throws balls into air. He throws one whenever the previous one is at its highest point. If he throws n balls each second, the height to which each ball will rise is

- (a) $\frac{g}{2n^2}$ (b) $\frac{g}{n^2}$
(c) $\frac{2g}{2n}$ (d) $\frac{2g}{2n^2}$

Sol . (a)

Time taken by each ball to reach highest point,

$$t = \frac{1}{n} \text{ s}$$

As the juggler throws the second ball, when the first ball is at its highest point, so $v = 0$

Using $v = u + at$, we get

$$0 = u + (-g)\left(\frac{1}{n}\right)$$

$$\text{or } u = \left(\frac{g}{n}\right) \quad \dots(i)$$

$$\text{Also } v^2 = u^2 + 2aS$$

$$\therefore 0 = \left(\frac{g}{n}\right)^2 + 2(-g)h \quad \text{using (i)}$$

$$\text{or } h = \frac{g}{2n^2}$$

INTEGERS TYPE QUESTIONS

25. A cricketer hits a ball in a vertical x-y plane from the ground level with a velocity $\vec{v}_0 = (10\sqrt{3}\hat{i} + 30\hat{j})$ m/s. Find the time in which velocity vector makes an angle of 30° with horizontal x-axis. (Take $g = 10 \text{ m/s}^2$.)

Ans. 2 sec

26. A car accelerates from rest at a constant rate of 2 m/s^2 for some time. Then, it retards at a constant rate of 4 m/s^2 and comes to rest. What is the maximum speed attained by the car if it remains in motion for 3 seconds?

Sol. $v = a_1t_1 = a_2t_2$

$$\therefore t_1 = (a_2/a_1) t_2 = (4/2)t_2 = 2t_2.$$

$$\text{Next, } t_1 + t_2 = 3 \text{ or } 2t_2 + t_2 = 3t_2$$

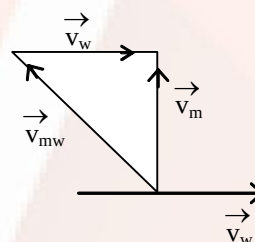
$$t_2 = 1 \text{ sec and } t_1 = 2 \text{ sec}$$

$$\therefore v = 2 \times 2 = 4 \text{ m/s.}$$

27. A swimmer wishes to reach directly opposite bank of a river, flowing with velocity 8 m/s. The swimmer can swim 10 m/s in still water. The width of the river is 480 m. Time taken by him to do so is

$$\text{Sol. Time taken} = \frac{d}{v_m} = \frac{d}{\sqrt{v_{mw}^2 - v_w^2}} = \frac{480}{\sqrt{10^2 - 8^2}} =$$

$$\frac{480}{6} = 80 \text{ s}$$



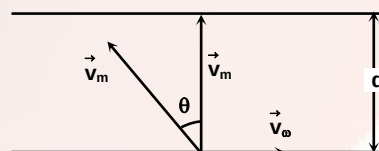
28. A boat which has a speed of 5 km/hr in still water crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water, in kilometer per hour, is

$$\text{Sol. } \vec{v}_m = \vec{v}_{m\omega} + \vec{v}_\omega$$

$$\vec{v}_{m\omega} = \vec{v}_m - \vec{v}_\omega$$

$$v_{m\omega}^2 = v_m^2 + v_\omega^2 - 2 v_m v_\omega \cos 90^\circ$$

$$v_{m\omega}^2 = v_m^2 + v_\omega^2$$



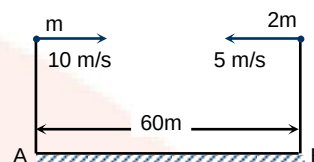
No

$$\frac{\omega_{m\omega}^2}{d^2} = \frac{v_m^2}{d^2} + \frac{v_\omega^2}{d^2}$$

$$\Rightarrow \frac{25}{1} = \left(\frac{60}{15}\right)^2 + \frac{v_\omega^2}{1}$$

$$\Rightarrow v_{\omega} = 3 \text{ km/hr}$$

29. Two particles, one of mass m and the other of mass $2m$, are projected horizontally towards each other from the same level above the ground with velocities 10 m/s and 5 m/s , respectively. They collide in air and stick to each other. The distance of the combined mass from point A is



Sol. They will fall through the same height say h . Let them collide at a distance x from tower A.

$$\therefore h = \frac{1}{2}gt^2 \quad \dots (1)$$

$$x = 5t \quad \dots (2)$$

$$60 - x = 10t \quad \dots (3)$$

solving (2) and (3) we get $x = 20 \text{ m}$, $t = 4 \text{ sec}$, $h = 80 \text{ m}$

30. A constant retardation of 2.5 m/s^2 is applied to a body of mass 20 kg moving initially with a speed of 15 m/s . How long does the body take to stop?

Sol. $v = u - gt$

$$t = u/g \text{ as } v = 0$$

$$= \frac{15}{2.5} = 6 \text{ sec}$$

